

1. Explain why the example results either match or do not match your results.

Because which thread is going to execute is unpredictable and either the producer threads might execute many times before the consumer ever executes or the consumer might execute as soon as one producer threads posts that there is an element in the buffer. Also the numbers are generated at random.

2. For this specific problem, what is the primary purpose of using semaphores?

The semaphores are used to synchronize access to the buffer.

3. For this specific problem, what is a semaphore's role in synchronization between producer and consumer threads?

The semaphores are used to synchronize access to the buffer. Specifically, they help in preventing a consumer thread try to delete an item from the buffer when it's empty and also prevent the producer thread from adding an item when the buffer is full.

4. Describe what happens when the producer fills the buffer. Under what circumstances can data be lost?

If the producer happens to fill the buffer any other producer thread trying to add an element in the buffer will be put to sleep by the `sem_wait` operation and will only run after the consumer thread signals that the buffer is now not full. Data can be lost if the buffer is overwritten by the producer before the consumer consumes that index.

5. Describe what happens when the buffer is empty?

If the buffer is empty then any other consumer threads will be put to sleep by the `sem_wait(&full)` operation, which waits for a producer thread to signal that the thread now has elements in it.